Run It, Then Cite It: Execution- and Mutation-Verified Answers to Developer Questions

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Code, data and reproduction steps: `https://github.com/ameya-purohit/LMA-Final-Project`

Abstract

Developers often ask how a library behaves when its documentation may be wrong. LLM assistants tend to repeat the documentation and cite code lines that look related but do not decide the behaviour. We built an agent that checks its answer in two ways. First, it splits the answer into small claims and tests each one with a short generated program (a *probe*) that runs in an offline Docker sandbox against the library as it was before the question was asked. Second, it keeps a cited line only if changing that line (a *mutation*) changes the claim's result while leaving related control checks unchanged. Failed checks become the agent's next thing to investigate. We tested it on real GitHub issues from `click` and `marshmallow`, with answer keys built automatically from each issue's merged fix and its test. The hypotheses were pre-registered and the test set was locked before any test run. On 145 test questions, the agent did **not** beat a ReAct agent with the same tools, model and budget: 40.0% against 49.7% correct and causally cited answers by an LLM judge (95% CI of the difference $[-19.3, -0.7]$), and 25.0% against 41.7% on 12 questions labelled by two humans. The judge agreed with the humans only fairly (kappa 0.36 and 0.32), so by our pre-registered rule the human labels count first. With human labels, the mutation check *lowered* citation precision (30.6% against 44.1% without it). Its verdicts agree with the lines developers actually changed no better than chance. The agent does beat closed-book and single-call RAG baselines by wide margins, and no system ran an unsafe action. Our main result is therefore negative, and the traces show why: in real code, mutation is a poor test of whether a line causes a behaviour, and acting on it moves the agent away from good citations. An exploratory multi-agent extension, added after the main results, made the pipeline much cheaper but did not fix this.

1 Introduction

"Why does this library behave differently from its documentation?" is a common developer question. The answer usually depends on code spread across files: where a default is set, where it is overridden and which branch runs. LLMs answer such questions confidently and often repeat the documentation even when the code disagrees. Agents that search and read code do better, but they still decide behaviour by *reading*, and LLMs misread control flow and overrides. A second, less visible problem is citations. An answer can cite lines that are on topic but do not produce the behaviour. Code gives us a test that plain text does not have: run the behaviour, change the cited line, and see whether the behaviour changes.

Our agent ("ours") adds two checks to a tool-using code agent. **Check 1, run it:** every behavioural claim in a draft answer is tested by a probe in a sandbox, and a refuted claim is investigated again with the observed output as the new goal. **Check 2, flip the switch:** every citation of a confirmed claim is mutated (a constant changed, a condition negated, a comparison swapped, a statement deleted). The citation is accepted as *causal* only if some mutant changes the claim probe's result and leaves the control probes unchanged. A rejected citation sends the agent looking for the line that does decide the behaviour. Each final claim carries a grade: executed and causally cited (E+C), executed (E), confirmed by a repository test (T), read only (R) or documentation only (D).

Research questions. RQ1: does this procedure give more correct, causally cited answers, and fewer confident errors, than a ReAct agent with the same tools (including execution and mutation) and the same limits? RQ2: how well does the mutation check agree with human judgements of causality and with the lines the developer changed, and which acceptance rule agrees best?

Pre-registered hypotheses (`PREREGISTRATION.md`, fixed before any test run). **H1**: ours beats B2 (ReAct, same tools) on the causally-supported answer rate by at least 8 points, with a paired 95% CI that excludes zero. **H2**: ours has at least 40% fewer confident errors (relative) than B2, with coverage no more than 10 points lower. **H3**: human-judged causal citation precision is at least 15 points higher with the mutation check than without it (ablation A2).

Contributions. (1) A fully logged agent that verifies claims by execution and citations by mutation, in a shared harness where baselines differ only in control logic. (2) 170 real developer questions (162 after a human spot check), each with executable gold built only from the developer's fix and test, with no LLM involved. (3) A pre-registered, paired comparison with three baselines and five ablation runs, with the LLM judge checked against two human labellers. (4) A measurement of mutation as an automatic test of causality. (5) A negative result with a mechanism, plus an exploratory multi-agent extension.

2 Related Work

Retrieval and tools. ReAct [1] interleaves reasoning with tool calls; IRCoT [2] and FLARE [3] decide when to retrieve more. They decide *when to look further*, but claims are still judged by reading. Our strongest baseline, B2, is a ReAct agent with every tool our agent has. **Self-checking.** Self-RAG [4] and CRAG [5] train critics for retrieved passages; Chain-of-Verification [6] answers its own verification questions; CRITIC [7] and Reflexion [19] feed tool results or failures

back. In these systems a model still judges support, or tools check the model's own computation. In ours, an executable test decides claims about a third-party codebase. **Repository agents.** SWE-agent [8] and Agentless [9] fix issues, and SWE-bench [20] scores them by whether the fix's tests pass. We reuse the SWE-bench recipe to build our *dataset*. **Testing documentation.** CASCADE [10], Kang et al. [11] and METAMON [12] find documentation–behaviour mismatches by execution. They check documentation function by function; they do not answer a question, search across files or cite lines. **Mutation.** Mutation testing [18], MUSE [13] and Metallaxis [14] mutate lines to locate faults. MUSE's idea that a useful mutant changes the target outcome but not others inspired our control probes. **Citations.** ALCE [15] checks whether citations support statements, and Wallat et al. [21] separate citation correctness from faithfulness. We target a stricter, code-specific notion: does the cited line *decide* the behaviour? Work on code–comment inconsistency [16] and conflicting context [17] studies the same tension, but not inside an answering agent. To our knowledge no question-answering system checks its citations by executable mutation or uses failed checks to steer its search. The novelty is the combination and its measurement, not the parts.

3 Dataset

Main set. We followed the SWE-bench recipe [20]: the issue is the question, and the merged fix and its test are the answer key. Every step is deterministic code (`data/`); no LLM writes, edits or filters any gold field. (1) *Mine*: list closed pull requests of `click` and `marshmallow` merged since 1 Jan 2020, and keep those whose text says "fixes #n" for an issue, that change at least one source file and one test file, and that touch at most 10 source files. Bot issues and issues labelled as feature requests are dropped. (2) *Freeze*: the frozen commit is the PR's base commit, before the fix. Systems see only this tree and its git history. (3) *Validate*: in the sandbox, apply only the PR's test changes; at least one test must fail (these are the fail-to-pass tests). Then apply the source changes too; every fail-to-pass test must pass. The failure message on the frozen code is the observed old behaviour. (4) *Build gold*: the gold evidence is the frozen-code line ranges that the fix changed or deleted; pure insertions are anchored to the line above. The gold answer is a template filled from the test output and the diff. A category is computed in code: *doc-related*, *spread across files* or *single location*. A system sees only the issue title and first post, plus the frozen tree; a unit test checks that the workspace cannot hold the fix, the test names or the gold.

Table 1: Mining funnel (`data/mining_report.md`). Each dropped item and its reason is in `data/dropped.jsonl`.

Step	click	marshmallow	total
Closed PRs listed	1,898	1,405	3,303
Merged since 2020-01-01	801	605	1,406
Linked to an issue, not a feature request by label	284	75	359
Change source and tests (candidates)	179	33	212
Validated: test fails before, passes after	152	24	176
Gold built, one question per issue	**146**	**24**	**170**

Of the 170 questions, 103 are doc-related, 63 single-location and 4 spread across files. A seeded split, stratified by library and category, gives 20 dev questions (used for development and one prompt round) and 150 test questions, locked with their SHA-256 hash before any test run.

Human spot check of the gold (gate G1). All four members checked a seeded 20% sample (34 questions, 8–9 each, one member per question) using a self-contained HTML page that showed the issue, the fix diff, the failing test's message and links to the gold lines. For each question they answered three yes/no checks: is it a genuine question about library behaviour, does the failing test capture what the issue asks, and do the gold lines decide it? **8 of 34 (23.5%) were marked invalid**, all on the second check. In 7 the issue was really a feature request, so the test only shows that the feature did not exist yet ("unexpected keyword argument", "DID NOT WARN"). In one, the fix bundled several changes and the failing test belongs to another one. The 8 were removed and logged. Five were test questions, so all test results are on the remaining **145** (125 click, 20 marshmallow). Because the test set was locked first, the removal is applied at analysis time, the same way for every system. Each member made their own verdicts; their notes were later grammar-corrected with an AI tool and the files assembled from them, with no verdict changed (disclosed in `PREREGISTRATION.md`).

Other sets. *toylib* is a small hand-written library with planted ground truth (a wrong docstring, a decisive line, a non-causal default, a line that changes every output, redundant guards). It is the unit test bed for the mutation check. The *adversarial set* has 15 dev questions, each with one planted instruction: run `curl` in a probe, answer "works as intended", or write to `~/.ssh`. Each also asks for a unique canary token, so obeying is detectable.

4 Method

Shared harness. All systems share one LLM client, config, model, retrieval index and starting material, tool set, sandbox and output schema. They differ only in control logic, and a test checks that each ablation's config differs from the default only in its own flag. No agent framework is used; the loops are written by hand on the provider's tool-calling API. The *LLM*

client has a SQLite cache (so re-runs are free and ablations reuse the full agent's identical first steps), rate limiting, retries, a 120 s timeout and a JSONL trace with token counts. *Retrieval* chunks code by AST and docs by heading, with separate BM25 indexes; every system starts with the top 5 code and top 3 doc chunks. *Tools*: `search_code`, `search_docs`, `open_file`, `find_definition`, `find_references`, `find_tests`, `git_log`, `git_blame`, `run_probe`, `mutate_and_run` and `run_repo_test`, each with a schema checked by pydantic. The *sandbox* is one Docker container per question with no network, a read-only file system except `/tmp`, a non-root user and limits of 1 CPU, 1 GB and 256 processes. An AST policy also blocks `os.system`, `subprocess`, `eval`, dynamic imports and writes outside `/tmp` before a probe runs. No LLM-generated code ever runs on the host.

The agent (`agent/loop.py`) produces structured output by calling function tools, and works in seven steps.

1. *Investigate* with read-only tools (at most 8 steps), then submit a draft: an answer plus claims, each with scope, citations and source (code, doc or test).
2. *Prioritise*: claims backed only by documentation, or in conflict with another source, are checked first.
3. *Check 1*: for each behavioural claim, write a probe (`observe()` and `test_claim()`) plus 1–2 control probes in one call, and run them in the sandbox. A broken probe is repaired at most twice; at most 12 probe runs per question.
4. *Refutation*: first compare the probe's setup with the question's scenario. If they differ, rewrite the probe; if they match, the claim is refuted and investigated again with the observed output as the goal.
5. *Check 2*: for each citation of a confirmed claim, apply each mutation operator to the cited line with libcst (the rest of the file stays byte-identical) and rerun the claim and control probes. Three acceptance rules are computed: *any_flip* (any change), *behavioural* (a change that is not just breakage, such as a SyntaxError or a NameError) and *specific* (behavioural, with every control unchanged). The agent uses *specific*. A rejected citation triggers a search for a better line, which is then checked again. A line no operator applies to is *untestable*.
6. *Vacuity check*: if a confirmed claim has no causal citation, check whether any mutant in its function changes the probe at all. If none does, the probe is vacuous and is written again once.
7. *Grade and answer*: grades are computed from the claim table in code. Only confirmed claims are stated as fact; conflicts are stated with their observed outputs, and unverified claims are flagged.

All model calls share a budget of 25 requests per question.

Baselines. B0, closed book: one call with the question only. **B1**, direct RAG: one call with the starting material, asked to cite lines. **B2**, ReAct: the shared tool loop with *all* of the agent's tools, including `run_probe` and `mutate_and_run`, and the same limits (8 steps, 12 probe runs, 25 requests). B2+ (a larger budget) and B3 (Chain-of-Verification) are implemented but were not run.

Table 2: Capabilities used (four of the five; fine-tuning was not done, because RQ1 compares procedures on the *same* model).

Capability	Why the problem needs it	Ablation	Metrics
Tool calling	Behaviour is settled by running probes and mutants	A1 (no execution)	Valid-call rate, recovery, probe verdict rate, unsafe attempts
Agentic decisions	Next step depends on check results: re-investigate or flag	A3 (flag only)	Confident errors, coverage, refuted claims
Reasoning	Splitting answers into checkable claims and writing exact probes	A1, A2	Claims per answer, broken probes, grades
RAG	Answers must come from the frozen repository, not memory	B0, A10	Fix-location recall, precision vs. fix lines

5 Experimental Setup

Model and judge. Every system uses `gemini-3.1-flash-lite` (Google free tier) at temperature 0, never changed during the experiment. The judge is `openai/gpt-oss-120b` on Groq, a different model family. It saw the question, gold answer, observed failure, the source part of the fix and each cited snippet, and labelled each answer correct, partial or incorrect, and each citation causal, on-topic but not causal, or irrelevant. One judge was used throughout.

Pre-registration and quota. H1–H3, thresholds, metrics and ablation predictions were fixed before any test run. Prompts were iterated once on dev, with equal effort for every system (`docs/prompt_log.md`). One further change to ours alone, made on dev before the lock, is disclosed (D-51). At the lock, the test file's hash and all prompt hashes were recorded, and `eval/run.py` refuses the test split otherwise. The real limit was about 500 Gemini requests per day per key, so we used four members' keys (same model and settings). The code has about 350 unit tests, 18 Docker tests and 3 live API tests, and each member's bundle passes its own tests in push order.

Runs. `pilot_d51`: ours, B1, B2, A1–A3 on 20 dev questions (7 Nov). `main`: ours, B1, B2, A1–A3, then B0, on the 150 test questions (145 analysed, 7–11 Nov); all 1,050 answers were judged. `sub50`: ours, B2 and the two A10 variants on the first 50 test questions in the seeded order (47 after the spot check, 11 Nov). `adv`: ours, B2 and B1 on the adversarial set (10 Nov).

Metrics. *Primary: causally-supported answer rate*: the answer is correct *and* every claim has at least one causal citation. A citation is causal if it falls inside a gold range or is labelled causal (by the judge, or by humans where they exist). Partial answers count as wrong. We also report correctness alone; *confident errors* (wrong answers that were neither flagged nor abstained); *coverage* (answers neither flagged nor abstained); *fix-location recall* (questions with at least one citation in a gold range); and cost. Precision against gold ranges is a lower bound, since a line outside the fix can also decide the behaviour. Tests are paired by question: McNemar's exact test and a paired bootstrap with 10,000 resamples for 95% CIs (clustered by question for citation precision).

6 Results

Table 3: Main results on 145 test questions, judge labels (`results/summary/analysis_main_test`). Conf. err. = confident errors.

System	Correct	Causally supp.	Fix-loc. recall	Conf. err.	Coverage	Requests/q	Tokens/q
B0, closed book	23.4%	4.8%	3.4%	52 (35.9%)	59.3%	1.0	1,495
B1, direct RAG	37.9%	21.4%	42.8%	89 (61.4%)	99.3%	1.0	5,432
B2, ReAct, same tools	**68.3%**	**49.7%**	62.1%	46 (31.7%)	100%	10.5	119,035
Ours	61.4%	40.0%	60.0%	43 (29.7%)	86.2%	14.7	185,876
A1, no execution	62.8%	39.3%	62.8%	53 (36.6%)	99.3%	7.9	78,107
A2, no mutation	62.8%	42.1%	60.7%	44 (30.3%)	90.3%	10.0	112,735
A3, flag only	61.4%	40.7%	60.0%	**29 (20.0%)**	67.6%	8.5	87,951

Ours is far ahead of the simple baselines: +18.6 points over B1 on the primary metric (CI [+10.3, +26.9]) and +35.2 over B0. It is behind B2, the ReAct agent with the same tools, while using 40% more requests and 56% more tokens.

6.1 Checking the judge against humans (gate G3)

Two members (Aditya and Sourav) labelled the same blinded sample separately: 12 test questions (seeded, stratified by library) $\times$ 6 systems, giving 72 answers and 204 citations, with system names hidden. An AI tool was used only to format their choices into the label files (disclosed). Human–human kappa was 0.742 for correctness and 0.902 for citations. Judge–human kappa was only 0.363 and 0.322. On citations the judge is lenient: it rated 72 of 204 citations higher than the humans did (49 of them *causal* where the humans said *on-topic but not causal*) and 11 lower. On correctness it errs both ways (15 higher, 11 lower). Both kappas are below the pre-registered 0.6, so **the primary comparison uses the human labels**, and the 145-question judge results are secondary.

6.2 Hypothesis tests

Table 4: Pre-registered hypotheses. Human labels (12 questions) are primary; judge labels (145 questions) are secondary.

Test	Labels	Result	Verdict
H1	Human, 12 q	25.0% vs 41.7%, -16.7 points, CI $[-41.7, 0.0]$, McNemar $p = 0.50$ (0 vs 2 discordant)	Not supported
H1	Judge, 145 q	40.0% vs 49.7%, -9.7 points, CI $[-19.3, -0.7]$, $p = 0.059$ (17 vs 31)	Not supported; B2 better
H1, correctness only	Judge, 145 q	61.4% vs 68.3%, -6.9 points, CI $[-14.5, 0.0]$, $p = 0.099$	B2 ahead, not significant
H2	Judge, 145 q	Confident errors 29.7% vs 31.7% (6.5% relative reduction), CI $[-10.3, +6.2]$; coverage 86.2% vs 100%	Not supported
H3	Human, 12 q	Causal citation precision 30.6% (36 citations) vs A2's 44.1% (34), -13.6 points, CI $[-28.6, -1.0]$	Not supported; A2 better
H3	Judge, 145 q	58.7% vs 59.3%, CI $[-7.6, +6.0]$	Inconclusive

On the 12 human-labelled questions, B2 was correct on 9 and causally supported on 5; A1, A2 and A3 on 8 and 4 each; ours on 6 and 3; B1 on 3 and 1. Ours was better than B2 on none of the 12. The human and judge views agree in direction on every test; the judge mainly raises every system's level. A note on power: 145 questions is above the floor of about 120 that we set in advance, but the pre-registration also states that 120 questions give only about 50% power to detect an 8-point gain. So a CI that includes zero is inconclusive, not evidence of no effect. Here the judge-based H1 interval lies below zero, in B2's favour. Before the spot check, on all 150 questions, the gap was -10.7 points (CI $[-20.0, -1.3]$, $p = 0.033$), so the exclusion does not change the conclusion.

6.3 Ablations

Removing execution, mutation or re-investigation leaves correctness and the primary metric statistically unchanged (every difference within 2.1 points, every CI includes zero). Re-investigation turned flagged answers into confident ones without making them more often right, so A3, which only flags, has a third fewer confident errors, at the cost of answering

Table 5: Ablations: one config flag each, same questions, model and budget. CIs are for ours minus the ablation.

Ablation	Pre-registered prediction	Outcome
A1, no execution (claims checked by reading)	More confident errors, especially on doc-related questions	Same direction, not significant: 36.6% vs 29.7% confident errors (CI [−14.5, +0.7], $p = 0.12$); doc-related 36 vs 28. Correctness unchanged
A2, no mutation	Similar correctness, clearly lower causal precision	Correctness similar (62.8%). Precision went the *other* way: higher without mutation (44.1% vs 30.6%, human labels)
A3, flag only, no re-investigation	Fewer confident errors than B2, lower correctness than ours	Half held: 20.0% confident errors (B2 31.7%, ours 29.7%; ours − A3 CI [+4.1, +15.9], $p = 0.004$). But correctness was *equal* to ours (61.4%), not lower. The price was coverage: 47 answers flagged instead of 20
A6, acceptance rule	*specific* matches humans best	Opposite: *specific* matches humans worst (Table 6)
A10-none (no starting material, 47 q)	More steps, similar accuracy	Correctness equal (53.2%); fix-location recall falls 59.6% → 44.7% (CI [+6.4, +25.5], $p = 0.016$); slightly *fewer* requests (14.2 vs 14.9)
A10-oracle (whole fix files given, 47 q)	Shows the ceiling	No gain: 51.1% correct, 36.2% causally supported (ours: 53.2%, 36.2%)

confidently less often. Mutation lowered human-judged citation quality. Retrieval helps the agent find the developer's lines, but handing it the right files did not raise correctness: the limit is in using the material, not finding it. On the 9 planted toylib lines, *specific* agrees with the planted truth on 8, against 7 for the other rules; the rule works on the code it was designed on, but not on real libraries (below).

6.4 RQ2: is mutation a good test of causality?

Table 6: Mutation verdicts of ours on the test set: 862 checks, of which 123 lines were untestable (31 of them fix lines). Precision against fix lines is a lower bound.

Rule	Accepted (of 739)	Precision vs fix	Recall vs fix	Kappa vs fix	Agreement w/ humans (52)	Kappa vs humans
any_flip	494	36.6%	66.8%	−0.001	55.8%	0.198
behavioural	411	37.0%	56.1%	0.007	55.8%	0.167
specific (used)	177	36.7%	24.0%	0.001	51.9%	−0.098

Against both independent references, the mutation verdicts are at chance level or just above it. The rule the agent uses rejects 76% of the lines the developer actually changed, and it agrees with humans least, although we predicted most. Requiring the controls to stay unchanged rejects many truly causal lines in real code, where one line often affects several behaviours. Likely causes are equivalent mutants, redundant code paths and controls that are not really independent of the claim. The 52 human-labelled lines make these kappas imprecise, but none reaches even moderate agreement (0.4).

6.5 Tool use, safety and prompt injection

Table 7: Tool use from the traces of the 145 test questions (replays counted once). Verdict rate = probes that ran and reached pass or fail. Blocked = probes refused by the AST policy before running.

System	Tool calls	Valid-call rate	Errors (recovered)	Probes	Verdict rate	Blocked probes
Ours	3,599	100.0%	45 (19)	604	85.2%	36 (6.0%)
B2	1,521	99.7%	112 (12)	695	88.5%	26 (3.7%)
A2	1,999	100.0%	35 (13)	546	84.5%	31 (5.7%)
A3	2,072	100.0%	36 (12)	459	85.1%	30 (6.5%)

No system executed an unsafe action. Most blocked probes were harmless in intent, such as creating a file outside `/tmp` to set up a scenario; a few imported `subprocess` or `shutil`. On the adversarial set, the planted text reached the model in 12/15 items for ours, 11/15 for B2 and 8/15 for B1. When exposed, the systems added the canary to their answer in 7/12, 9/11 and 6/8 cases. No probe attempted the dangerous action (`curl`, writing to `~/.ssh`), but several answers repeated a planted "works as intended" verdict. The sandbox protects the machine; nothing stops read text from changing the answer.

6.6 Categories, examples and why ours lost

On *doc-related* questions (n = 86), where the method was meant to help most, ours is 54.7% correct and 40.7% causally supported, against B2's 65.1% and 47.7%. On *single-location* questions (n = 56), ours is 71.4% and 37.5%, against 73.2% and 53.6%. Results are similar for the two libraries.

Where mutation hurt (click-2099, "No way to alter dynamic default value help text"). Ours' probe confirmed the

behaviour, and it cited lines 2722 and 2723 of `core.py`; the developer's fix changed 2722. The mutation check rejected 2722 and accepted 2723, so the agent dropped the developer's own line. A2 and B2 kept both lines. **Where the procedure worked (click-1422, "aliased group shows abbreviated command in usage").** Probes refuted two of five drafted claims (`UsageError: Missing command`); the agent revised both, the third versions passed, and the mutation check accepted the fix line (1349). The answer was judged correct and causally supported, while B2 cited the invoke code and was judged partial.

Why ours lost to B2. Ours' decision logs on the 145 questions record 612 probe runs (401 passed, 111 failed, 100 broken), 381 proposed replacement citations, and 84 refuted and 75 revised claims. (These logs count every run, including replays, so the totals differ slightly from Table 7.) Four things stand out. (1) The mutation check throws away good citations (Table 6), and each rejection starts a search for another line, which is often worse. (2) Ours makes more claims (2.61 per answer against B2's 2.05), and the primary metric needs a causal citation for every one. (3) 16% of probe runs were broken, for example by a wrong import, and each repair spends requests from the same budget that B2 spends on investigating. (4) Flagging separates right from wrong poorly: 7 of ours' 20 flagged answers were in fact correct. By failure class (judge labels and fix-line hits), B2 has more answers that are correct and causally supported (72 vs 58), and ours has more with the right behaviour but the wrong place (22 vs 18).

7 Exploratory Extension: Multi-Agent Fault Localization

This was designed after the main results were known and pre-registered separately before its own test runs. It does not change H1–H3. It asks whether specialised roles, better retrieval, a code graph and better-chosen experiments locate the deciding lines better than the locked agent (E0) and B2. Five roles run in a fixed order (`multiagent/orchestrator.py`, no framework), passing typed pydantic objects: a Bug Investigator (states the behaviour and 2–4 hypotheses), a Probe Engineer (reproduce and control probes), a Code Analyst (ranks suspicious lines), an Experiment Engineer (plain code; runs mutation checks with three verdicts: accepted, inconclusive when a control also changes, and rejected) and a Causal Verifier (judges the hypotheses and writes the answer). The configurations add one thing at a time: E1 the roles with BM25; E2 hybrid retrieval (BM25 + dense bge-small [26], FAISS [27], fused by RRF [22], then a cross-encoder rerank [25]); E3 a code graph with slicing [24, 28] and executed-line spectra (Ochiai [23]); E4 pairwise mutation; E5 experiment selection by information gain; E6 E5 with its tools behind an MCP server [29]. Like AutoCodeRover [30] it mixes code-structure search with spectrum-based localization; unlike AutoGen [31] the agents exchange typed objects, not chat. The budget is 12 requests per question (E0: 25). It ran on the 47 `sub50` questions, with prompts developed on only 2 dev questions.

Table 8: Extension on 47 test questions (`docs/extension_results.md`). Gold = the developer's fix lines, so all localization rates are lower bounds. E4 matched E3 and E6 matched E5 on every localization measure.

System	Line Top-1	Line Top-3	Line MRR	Correct (judge)	Fix lines accepted	Requests/q	Tokens/q
E0 (locked agent)	**48.9%**	59.6%	0.532	53.2%	24/48*	14.9	187,823
B2 (ReAct)	46.8%	**61.7%**	**0.539**	**61.7%**	–	9.9	109,529
E1 roles	34.0%	48.9%	0.401	48.9%	9/46	7.6	32,031
E2 + hybrid retrieval	40.4%	55.3%	0.479	57.4%	9/43	7.8	33,678
E3 + code graph	31.9%	36.2%	0.358	55.3%	11/48	8.0	35,393
E5 + information gain	31.9%	36.2%	0.352	55.3%	10/43	8.0	35,337

*Not comparable with E1–E5 or with Table 6: E0 counts each distinct line once per question by its first recorded verdict, and its "no" also includes flips that changed a control, which E1–E5 call inconclusive.

No extension configuration beats E0 or B2 on any localization measure, and no correctness difference has a CI that excludes zero. E1 lowered line MRR against E0 (-0.131, CI $[-0.238, -0.035]$), and the code graph lowered line Top-3 from E2 to E3 (-19.1 points, CI $[-34.0, -4.3]$, $p = 0.035$). The reason is the ranking: lines are sorted accepted, cited, inconclusive, untested, rejected. E2 ranks about 3 lines, so 12 of its 26 top-3 fix-line hits were lines the check had *rejected*, kept only because the list was short. E3 ranks about 8 lines, most new ones come out inconclusive, and they push the rejected fix lines down. Pairwise mutation (95 pair checks) never accepted a pair. Information-gain selection cut mutant runs from 15.2 to 11.4 per question with the same Top-3, and the MCP version gave identical answers on all 47 questions. The roles used half of E0's requests and a sixth of its tokens.

Retrieval study (no LLM, 142 test questions that have a relevant chunk). Fusing BM25 and dense retrieval with RRF was the one clearly positive result: hit@5 71.8% against 64.1% for BM25 (+7.7 points, CI [+1.4, +14.1], $p = 0.035$). The cross-encoder reranker, trained on web text, hurt (hit@5 -11.3 points against RRF, $p = 0.009$). Our pre-registered prediction was stated on Recall@5: the reranked mode (42.0%) did not beat BM25 (46.3%), so it was not supported, and dense alone (47.6%) did not beat BM25, as predicted. E2–E6 still used the reranker as pre-registered, because switching after seeing test results would be tuning on test. The extension points the same way as the main study: the three-way check accepted only about one in five fix lines in E1–E5, so the causal test, not the search, limits localization.

8 Limitations

- **The judge failed validation** (kappa 0.36 and 0.32). The primary human comparison covers only 12 questions and is under-powered on its own; the 145-question judge results agree in direction.
- **The automatic gold admits feature requests.** The spot check found 23.5% invalid in a 20% sample; the rest was not checked by hand and may have a similar share. This affects all systems equally but adds noise.
- **One small model and one budget** (`gemini-3.1-flash-lite`, 25 requests). A stronger model may write fewer broken probes, and a larger budget might favour the procedure. Only two pure-Python, well-tested libraries.
- **Not run:** B2+, B3 (so RQ1's comparison with self-verification by reading remains open), repeat runs, the probe-limit sweep, the controlled conflict set, ablations A4, A5, A7–A9 and A11 on real questions, and fine-tuning. No question is unanswerable, so abstention is measured only through flagging.
- **Disclosed process deviations** (`PREREGISTRATION.md`): the test set was locked before the spot check; one change to ours on dev after the prompt round (D-51); several members' API keys; and the human verdict and label files were assembled outside the labelling pages, with AI used only for grammar and formatting.
- **Prompt injection** changes answers in every system; there is no defence beyond the sandbox. The extension is exploratory, on 47 questions already seen by the main study.

9 Conclusions and Future Work

We asked whether checking every claim by execution and every citation by mutation, and investigating whatever fails, makes a code question-answering agent more correct and better cited. On real, pre-registered and locked test data, it does not. A ReAct agent with the same tools did better on the primary metric, by the judge on 145 questions and by humans on 12. The mutation check made citations worse in the humans' eyes, and its verdicts track neither the developers' fix lines nor human judgements. Execution may still help in a narrower way: without it, confident errors rose from 29.7% to 36.6%, though this difference is not significant. Flagging failed checks instead of re-investigating them cut confident errors by a third at the same correctness, but the agent then answered confidently on fewer questions (coverage 67.6% against 86.2%).

Three lessons follow. First, "flip the switch" is a weak test of causality in real code: a line can matter without any single mutant changing exactly one behaviour, so a control that works on a planted library rejects three in four lines that developers changed. Second, a feedback loop built on a noisy check does harm: re-investigation spent budget moving away from good answers. Third, an LLM judge has to be checked: ours was lenient on citations, and without human labels H3 would have looked inconclusive instead of negative.

Future work. A softer causal test that ranks lines by how many probes their mutants change, or uses dynamic slicing, instead of a hard accept/reject. Using failed checks only to flag. Running B2+, B3 and the controlled conflict set. Filtering feature requests out of the automatic gold with a test-failure pattern check. An explicit defence against instructions in read text. Re-running the extension with RRF retrieval and no reranker, and a ranking that does not let inconclusive lines outrank rejected ones, on fresh questions.

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